Customer Segmentation Analysis Using SQL

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1. Project Overview

Objective

Businesses often have thousands of customers, but not all customers contribute equally to revenue. Identifying **high-value customers** helps companies design targeted marketing strategies, improve retention, and increase profitability.

This project performs **customer segmentation using SQL** by analyzing purchasing behavior such as **purchase frequency, spending patterns, and recency of transactions**.

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2. Business Problem

The company wants to answer the following questions:

- Which customers generate the **highest revenue**?
- Which customers purchase **most frequently**?
- Which customers have **stopped purchasing (churn risk)**?
- How can customers be segmented for **targeted marketing strategies**?

The goal is to classify customers into meaningful segments such as:

- High-value customers
- Loyal customers
- Occasional buyers
- At-risk customers

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3. Dataset Description

The dataset contains **transaction-level sales data**.

Column	Description
order_id	Unique identifier for each order
customer_id	Unique identifier for each customer
order_date	Date of transaction
product_id	Product identifier
quantity	Number of items purchased
price	Price per unit
total_amount	Total value of the order

Example dataset size:

- **150,000 transactions**
- **5,000 customers**
- **2 years of sales data**

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4. Tools & Technologies

- **SQL Server**
- **T-SQL**
- **Window Functions**
- **Aggregations**
- **CTEs (Common Table Expressions)**

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5. Project Workflow

The analysis follows these steps:

1. Data Cleaning
2. Check for NULL values
3. Check for Duplicate values
4. Customer Revenue Analysis
5. Purchase Frequency Analysis
6. Recency Analysis
7. Average Order Value
8. Customer Lifetime Value
9. Revenue Ranking Using Window Function
10. Customer Segmentation
11. Pareto Analysis (80/20 Principle)
12. Business Insights

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6. Data Cleaning

Before analysis, the dataset was cleaned to ensure accuracy.

Key steps:

- Removed **duplicate orders**
- Filtered **invalid transactions**
- Ensured **date formats were correct**

This ensures only valid transactions are analyzed.

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7. Revenue Analysis

First, we calculate **total revenue generated by each customer**.

```
SELECT  customer_id,
        ROUND(SUM(total_amount), 2) AS total_revenue
FROM orders
GROUP BY customer_id
ORDER BY total_revenue DESC ;
```

Purpose: Identify **high-spending customers**.

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8. Purchase Frequency Analysis

Next, we determine how frequently customers place orders.

```
SELECT    customer_id,
          COUNT(order_id) AS purchase_frequency
FROM orders
GROUP BY customer_id
ORDER BY purchase_frequency DESC;
```

Purpose: Identify **loyal customers who purchase frequently**.

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9. Recency Analysis

Recency measures how recently a customer made a purchase.

```
SELECT  customer_id,
        MAX(order_date) AS last_purchase_date,
        DATEDIFF(day, MAX(order_date), GETDATE()) AS days_since_last_purchase
FROM orders
GROUP BY customer_id
ORDER BY days_since_last_purchase DESC;
```

Purpose: Identify customers who may be **at risk of churning**.

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10. Average Order Value

Average order value placed by customers.

```
SELECT customer_id,
        ROUND(AVG(total_amount), 2) AS avg_order_value
FROM orders
GROUP BY customer_id
ORDER BY avg_order_value DESC;
```

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11. Customer Lifetime Value (CLV)

Customer Lifetime Value estimates the total value a customer brings.

```
SELECT  customer_id,
        ROUND(SUM(total_amount), 2) AS lifetime_value
FROM orders
GROUP BY customer_id
ORDER BY lifetime_value DESC;
```

Purpose: Understand **which customers contribute the most revenue**.

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12. Revenue Ranking Using Window Function

```
SELECT  customer_id,
        ROUND(SUM(total_amount),2) AS total_revenue,
        RANK() OVER(ORDER BY SUM(total_amount) DESC) AS revenue_rank
FROM orders
GROUP BY customer_id;
```

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13. Customer Segmentation

Customers are segmented based on **RFM analysis**:

- **Recency** – How recently a customer purchased
- **Frequency** – How often they purchase
- **Monetary Value** – How much they spend

Example segmentation query:

```
WITH customer_metrics AS(
SELECT
        customer_id,
        ROUND(SUM(total_amount),2) AS Monetary,
        COUNT(order_id) AS Frequency,
        DATEDIFF(day, MAX(order_date), GETDATE()) AS Recency
FROM orders
GROUP BY customer_id
)
SELECT *,
CASE
        WHEN Monetary > 5000 AND Frequency > 10 THEN 'High Value Customers'
        WHEN Frequency > 8 THEN 'Loyal Customers'
        WHEN Recency > 180 THEN 'Churn Risk'
ELSE 'Regular Customers'
END AS customer_segments
From customer_metrics;
```

Purpose: Group customers into **actionable marketing segments**.

Pareto Analysis (80/20 Principle)

```
WITH revenue_table AS (
SELECT customer_id,
          ROUND(SUM(total_amount),2) AS revenue
FROM orders
GROUP BY customer_id
),
revenue_calc AS (
SELECT customer_id,
          revenue,
          SUM(revenue) OVER() AS total_revenue,
          revenue * 100.0 / SUM(revenue) OVER() AS revenue_percentage
FROM revenue_table
```

```
)
SELECT * FROM revenue_calc
ORDER BY revenue DESC;
```

15. Key Insights

The analysis revealed several important insights:

- A small group of customers contributes **a large portion of total revenue**.
- **High-frequency customers** represent the most loyal segment.
- Some customers have not purchased in **over 6 months**, indicating potential churn risk.
- Targeted marketing campaigns could significantly increase retention among **high-value customers**.

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16. Business Recommendations

Based on the analysis:

- Offer **loyalty rewards** for high-value customers.
- Launch **re-engagement campaigns** for inactive customers.
- Provide **personalized product recommendations** to frequent buyers.
- Focus marketing budgets on **segments with the highest revenue potential**.

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17. Conclusion

This project demonstrates how SQL can be used to analyze customer purchasing behavior and generate meaningful customer segments. These insights can help businesses improve **customer retention, marketing effectiveness, and revenue growth**.